

Defence Institute of Advanced Technology (DU)
Department of Applied Mathematics

AM608: Data Structures and Algorithms

Assignment – 1

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Question 1

Q1. Find the value of c and n_0 such that $f(n) = \sqrt{n}$ is $\Omega(\log n)$.

Solution. We need to show $\exists c > 0, n_0$ such that for all $n \geq n_0$,

$$\sqrt{n} \geq c \log n.$$

Consider the function $h(n) = \frac{\sqrt{n}}{\log n}$ for $n > 1$. As $n \rightarrow \infty$, $h(n) \rightarrow \infty$, so there exists n_0 such that $h(n) \geq c$ for any fixed c (choose e.g. $c = 1$). Hence take $c = 1$ and pick n_0 sufficiently large (e.g. $n_0 = 16$ works since $\sqrt{16} = 4$ and $\log 16 \approx 2.77$). Thus $\sqrt{n} = \Omega(\log n)$.

Final choice: $c = 1$ and $n_0 = 16$ (one convenient choice). Hence $\sqrt{n} \geq \log n$ for all $n \geq 16$.

Question 2

Q2. Show constants c_1, c_2, n_0 such that $f(n) = 10n^2 - 3n$ is $\Theta(n^2)$.

Solution.

Proof. For $n \geq 1$ we have

$$10n^2 - 3n \leq 10n^2,$$

and for $n \geq 1$,

$$10n^2 - 3n \geq 10n^2 - 3n^2 = 7n^2.$$

So take $c_1 = 7$, $c_2 = 10$, and $n_0 = 1$. Thus $7n^2 \leq 10n^2 - 3n \leq 10n^2$ for all $n \geq 1$. □

$c_1 = 7$, $c_2 = 10$, $n_0 = 1$, so $10n^2 - 3n = \Theta(n^2)$.

Exercises from the Book

Q8. (R-3.2, R-3.3, R-3.6 – R-3.21).

Write each exercise heading and your solution. For long exercise lists, use `enumerate`:

1. R-3.2.

(statement)

Solution: ...

2. R-3.3.

Solution: ...

A Extra derivations

Here you can put extended algebra or scanned figures.